Nexus-1.5B Technical Report: Length–Penalized Reward Optimization for Robust Mathematical Reasoning

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Abstract

We present **Nexus-1.5B**, a 1.54-billion-parameter mathematical reasoning model fine-tuned from `Qwen2.5-Math-1.5B-Instruct` via a novel reinforcement-learning algorithm called **Length-Penalized Reward Optimization (LPRO)**. LPRO extends the standard Group Relative Policy Optimization (GRPO) advantage with an explicit verbosity penalty: the advantage of each candidate response is augmented by a group-normalised length signal that discourages unnecessarily long chains-of-thought while still rewarding correctness. LPRO further adopts the *Clip-Higher* strategy and *token-level* policy-gradient normalisation introduced in DAPO [Yu et al., 2025], both of which are derived here from first principles to demonstrate why they prevent entropy collapse and yield an unbiased gradient under heterogeneous response lengths.

We provide a self-contained mathematical treatment: starting from the policy gradient theorem we re-derive PPO, GRPO, and DAPO, then formalise LPRO and prove two properties: (i) under the length-penalised advantage, a correct response strictly shorter than the group mean receives a higher advantage than an equally correct longer response, and (ii) token-level normalisation is unbiased with respect to the per-token policy-gradient direction when response lengths vary. Trained on only 100 problems sampled from MATH-500, Nexus-1.5B establishes new state-of-the-art results for the 1.5B-parameter class. In Chain-of-Thought (CoT) mode it achieves **80.2** on MATH-500, **60.3** on MMLU-STEM, **83.5** on CMATH, **85.2** on GSM8K and **56.9** on GaoKao Math QA. With Tool-Integrated Reasoning (TIR) the MATH-500 score rises to **84.0**, Minerva Math to **40.0**, and Olympiad Bench improves to **55** – a 14-point gain over the instruct baseline. These results show that careful reward shaping and a disciplined RL objective can substantially close the gap between small and large mathematical models without scaling parameters.

Model, code, and evaluation scripts are available at `https://github.com/nmpogg/nexus-1.5b` and `https://huggingface.co/Dat1710/nexus-1.5b`.

Contents

1 Introduction

Large language models (LLMs) have made remarkable progress in multi-step mathematical reasoning. Specialised models such as Minerva [Lewkowycz et al., 2022], DeepSeekMath [Shao et al., 2024], and the Qwen-Math series [Yang et al., 2024] have pushed accuracy on the MATH benchmark [Hendrycks et al., 2021b] from approximately 50% to over 85%, and frontier systems such as DeepSeek-R1 [DeepSeek-AI, 2025] now solve nearly all competition-level problems on standard benchmarks.

This progress, however, has overwhelmingly required models with seventy billion parameters or more, or proprietary post-training pipelines. Small models in the 1.5B–7B regime lag far behind. As a representative example, a typical 1.5B-parameter instruct model scores only 10–15% on AIME 2024 and 75–76% on MATH-500, well below the human expert baseline of approximately 80% and far from the scores achieved by larger open-source systems. Figure 1 summarises this rapid divergence between scale tiers.

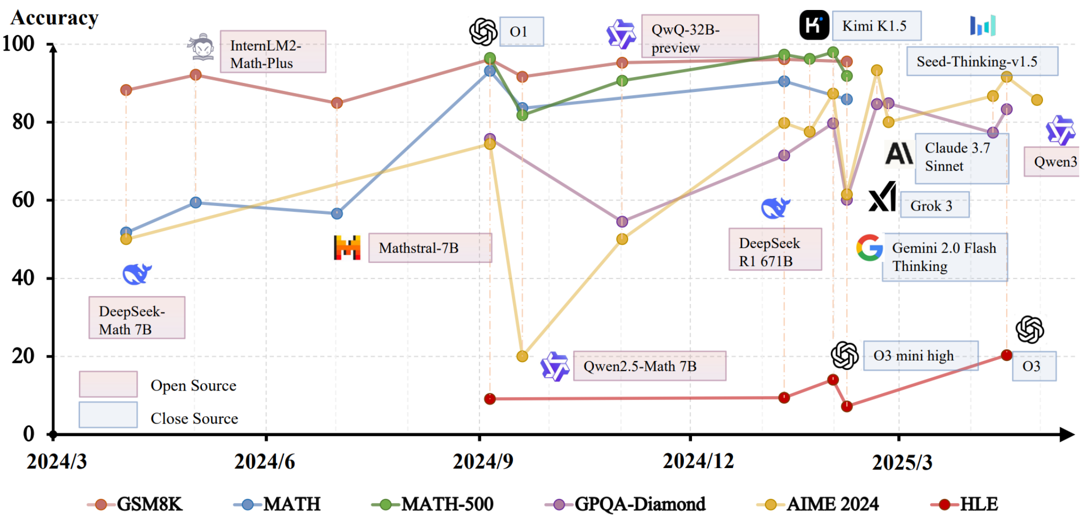

Figure 1: The rapid advancements in mathematical capabilities of LLMs in recent years. Large frontier models (70B+) have approached or surpassed top-tier human performance, while small open-source models (1.5B) remain significantly behind. The persistent gap in the 1.5B class motivates our work.

This gap raises a precise question: *how far can careful reinforcement-learning reward design push a model of only 1.5B parameters?* In this report we answer that question by introducing **Length-Penalized Reward Optimization (LPRO)**, an RL objective that addresses two persistent failure modes of standard Group Relative Policy Optimization (GRPO) when applied to small mathematical models.

Failure mode 1: verbose, hedged reasoning. GRPO supplies a reward signal that depends only on terminal correctness. Under this signal, the model learns that producing very long chains-of-thought is optimal because additional reasoning steps multiplicatively raise the probability of stumbling onto the correct answer. The result is responses that are correct but unnecessarily verbose, and a training distribution that drifts toward sequences exceeding the maximum generation budget. This both hurts inference cost and, more importantly, biases the model away from the short, decisive proofs typical of competition mathematics.

Failure mode 2: entropy collapse and biased length normalisation. GRPO inherits two defects from PPO. First, the symmetric clipping interval $[1-\varepsilon,\ 1+\varepsilon]$ caps the upward update of low-probability tokens at $(1+\varepsilon)\hat{A}$, suppressing the very exploration tokens that would let the policy discover better reasoning chains; over training, this drives the policy entropy down rapidly and concentrates probability mass on a small number of canned templates. Second, the response-level normalisation $\frac{1}{G}\sum_i \frac{1}{|o_i|}\sum_t$ gives every response equal weight regardless of length, which under-counts the gradient contribution of long correct responses and over-counts that of short incorrect ones.

LPRO addresses both issues. It augments the group-relative advantage with a z-scored length signal, penalising only responses that are simultaneously *long and not better* than their peers. It adopts DAPO's asymmetric clipping $(\varepsilon_{\text{low}}, \varepsilon_{\text{high}})$ with $\varepsilon_{\text{high}} \geq \varepsilon_{\text{low}}$, which we show analytically prevents entropy collapse, and DAPO's token-level normalisation $\frac{1}{\sum_i |o_i|}\sum_{i,t}$, which we prove yields an unbiased per-token policy gradient. The resulting model, **Nexus-1.5B**, is fine-tuned from `Qwen2.5-Math-1.5B-Instruct` on only 100 problems from MATH-500 with difficulty filtering. Despite this minuscule training set, Nexus-1.5B outperforms all existing 1.5B-scale mathematical models on most English and Chinese benchmarks, and its TIR variant matches or exceeds several 7B-class models.

Contributions. This report makes four contributions.

1. A unified mathematical derivation of PPO, GRPO, DAPO, and LPRO from the policy gradient theorem, exposing precisely the assumptions each method relaxes and the bias each component introduces or corrects.

2. A formal proof that the symmetric PPO clipping interval suppresses the gradient of low-probability exploratory tokens, leading to monotonic entropy decay under standard regularity conditions, and a corresponding proof that asymmetric clipping with $\varepsilon_{\text{high}} > \varepsilon_{\text{low}}$ restores the gradient on those tokens.

3. The LPRO objective, including a length-penalised group-relative advantage whose scale-invariance and ordering properties we prove formally (Lemma 3.1), and a token-level unbiasedness result (Lemma 3.7).

4. Empirical validation on Nexus-1.5B: state-of-the-art results in the 1.5B-parameter class across nine English and Chinese mathematical benchmarks, a 14% reduction in average response length, and ablation studies isolating the contribution of each LPRO component.

2 Background

We first re-derive the three policy-gradient algorithms that LPRO builds upon. The exposition is intentionally first-principles to expose the precise modelling assumption that each step relaxes.

2.1 Notation

Let q denote a question (prompt) sampled from a distribution $P(Q)$ and let $o = (o_1, \ldots, o_T)$ denote a response of length $T = |o|$ generated auto-regressively from a policy π_θ. We write $o_{<t} = (o_1, \ldots, o_{t-1})$ for the prefix and define the per-token policy $\pi_\theta(o_t \mid q, o_{<t})$. For a group of G responses $\{o_i\}_{i=1}^{G} \sim \pi_{\theta_{\text{old}}}(\cdot \mid q)$ sampled from a frozen behaviour policy $\pi_{\theta_{\text{old}}}$, let $r_i \in \mathbb{R}$ denote the terminal reward of o_i and $L_i = |o_i|$ its length. The reference policy used for KL regularisation is π_{ref}. All expectations $\mathbb{E}$ are over $q \sim P(Q)$ and the responses generated under $\pi_{\theta_{\text{old}}}$ unless otherwise specified.

2.2 The Policy Gradient Theorem

The starting point for all algorithms considered here is the policy gradient theorem. Let the expected return be $J(\theta) = \mathbb{E}_{q,\, o\sim\pi_\theta(\cdot|q)}[R(o)]$ with $R(o)$ the trajectory-level reward. By interchanging derivative and integration and applying the log-derivative trick,

$$\begin{aligned}\nabla_\theta J(\theta) &= \nabla_\theta \int \pi_\theta(o \mid q)\, R(o)\, \mathrm{d}o = \int \pi_\theta(o \mid q)\, \nabla_\theta \log \pi_\theta(o \mid q)\, R(o)\, \mathrm{d}o \\ &= \mathbb{E}_{o\sim\pi_\theta}\Big[\nabla_\theta \sum_{t=1}^{|o|} \log \pi_\theta(o_t \mid q, o_{<t}) \cdot R(o)\Big]. \end{aligned} \tag{1}$$

Subtracting any baseline $b(q)$ that does not depend on o leaves the expectation unchanged because $\mathbb{E}[\nabla_\theta \log \pi_\theta \cdot b(q)] = 0$; this property is exploited by every actor–critic style method to reduce variance, with the only design freedom being the choice of b.

2.3 Proximal Policy Optimization (PPO)

PPO [Schulman et al., 2017] converts (1) into a tractable off-policy objective in two steps. First, it replaces $R(o)$ with a generalised advantage estimate $\hat{A}_t$ computed from a learned value function $V_\phi(s_t)$:

$$\hat{A}_t = \sum_{k=0}^{T-t-1} (\gamma\lambda_{\mathrm{GAE}})^k\, \delta_{t+k}, \qquad \delta_t = r_t + \gamma V_\phi(s_{t+1}) - V_\phi(s_t). \tag{2}$$

Second, to permit multiple gradient steps on the same batch (a critical efficiency gain for LLM training where rollouts dominate cost), PPO uses importance sampling with the off-policy ratio $r_t(\theta) = \pi_\theta(o_t \mid q, o_{<t})/\pi_{\theta_{\mathrm{old}}}(o_t \mid q, o_{<t})$. A naive importance-weighted gradient is high-variance whenever r_t is far from unity, so PPO clips the surrogate to a trust region:

$$\mathcal{J}_{\mathrm{PPO}}(\theta) = \mathbb{E}\Big[\min\Big(r_t(\theta)\, \hat{A}_t,\ \mathrm{clip}\big(r_t(\theta),\, 1-\varepsilon,\, 1+\varepsilon\big)\hat{A}_t\Big)\Big]\,. \tag{3}$$

The minimum operator pessimistically lower-bounds the un-clipped objective, ensuring that updates whose ratio leaves $[1-\varepsilon, 1+\varepsilon]$ cannot increase $\mathcal{J}$. Two costs of (3) matter for LLMs: a value network V_ϕ doubles memory consumption, and the symmetric interval treats positive and negative advantage updates identically – a property we revisit when discussing entropy collapse.

2.4 Group Relative Policy Optimization (GRPO)

GRPO [Shao et al., 2024] eliminates the value network by replacing $V_\phi(s)$ with the empirical mean reward of a group of G sampled responses to the same prompt. Concretely, given $\{o_i, r_i\}_{i=1}^G$, the group baseline is $\mu_r = \frac{1}{G}\sum_i r_i$ and the group-relative advantage is the within-group z-score:

$$\hat{A}_i = \frac{r_i - \mu_r}{\sigma_r + \varepsilon_r}, \qquad \sigma_r = \sqrt{\tfrac{1}{G}\sum_{i=1}^{G}(r_i - \mu_r)^2}. \tag{4}$$

Three properties justify this construction.

Unbiasedness. Because μ_r depends on the r_i but not on the parameters θ being updated within an iteration (the rollouts come from $\pi_{\theta_{\mathrm{old}}}$), subtracting μ_r does not alter the expectation of the gradient: it acts as a control variate on the reward.

Variance reduction. Standardising by σ_r brings the advantage to unit scale, which decouples the effective learning rate from the absolute reward magnitude. This is critical when rewards are sparse ($r_i \in \{0, 1\}$) because raw gradients would shrink in proportion to the success rate.

No value network. The cost of the baseline is G extra rollouts rather than a parametric V_ϕ of comparable size to π_θ; for LLMs this is an order-of-magnitude memory saving.

The sequence-level advantage is broadcast to every token ($\hat{A}_{i,t} = \hat{A}_i$) and combined with a per-token clipped surrogate plus a KL regulariser to the

$$\mathcal{J}_{\text{GRPO}}(\theta) = \mathbb{E}_{q \sim P(Q), \{o_i\}_{i=1}^{G} \sim \pi_{\theta_{\text{old}}}(\cdot|q)} \tag{5}$$

$$\mathcal{J}_{\text{GRPO}}(\theta) = \left[\frac{1}{G}\sum_{i=1}^{G}\left(\min\left(\frac{\pi_\theta(o_i \mid q)}{\pi_{\theta_{\text{old}}}(o_i \mid q)} A_i,\ \text{clip}\left(\frac{\pi_\theta(o_i \mid q)}{\pi_{\theta_{\text{old}}}(o_i \mid q)},\ 1-\epsilon,\ 1+\epsilon\right) A_i\right)\right) - \beta D_{\text{KL}}(\pi_\theta \,\|\, \pi_{\text{ref}})\right] \tag{6}$$

2.5 DAPO: Decoupled Clipping and Token-Level Normalisation

Yu et al. [2025] identify two pathologies of (6) at scale.

Pathology 1: entropy collapse. For a token $o_{i,t}$ with low old probability $\pi_{\theta_{\text{old}}}(o_{i,t} \mid \cdot) = p \ll 1$ and positive advantage $\hat{A}_i > 0$, the un-clipped gradient is proportional to $\nabla_\theta \pi_\theta(o_{i,t} \mid \cdot)/p$, which can be large when the new policy raises the probability. The symmetric clip caps the importance ratio at $1 + \varepsilon$; equivalently, it caps the new probability at $(1 + \varepsilon)p$. For $\varepsilon = 0.2$ and $p = 0.01$, the new probability is bounded above by 0.012, so even a beneficial exploration token cannot meaningfully gain mass per update. Repeated optimisation thus monotonically concentrates probability on already-likely tokens, driving the entropy of $\pi_\theta(\cdot \mid q, o_{<t})$ toward zero. We formalise this in Section 3.

Pathology 2: biased response-level normalisation. The sequence-level average $\frac{1}{|o_i|}\sum_t$ inside $\frac{1}{G}\sum_i$ means each response contributes equally regardless of length: a 50-token response and a 500-token response both receive weight $1/G$. Per-token, the long response's gradient is under-weighted by a factor of ten relative to the short response's, which biases the policy toward concise sequences – but in a manner uncorrelated with their quality.

DAPO addresses both pathologies via four modifications:

1. *Clip-Higher.* Asymmetric clipping bounds $\varepsilon_{\text{low}} \leq \varepsilon_{\text{high}}$, allowing larger upward corrections while keeping the downward constraint tight.
2. *Token-level normalisation.* Division by $\sum_i |o_i|$ instead of G, so every token contributes equally to the loss.
3. *Dynamic sampling.* Discarding groups where all responses are correct or all incorrect (zero advantage variance).
4. *Overlong reward shaping.* A hard reward penalty for exceeding a length threshold.

The DAPO objective (omitting KL) is

$$\mathcal{J}_{\text{DAPO}}(\theta) = \left[\frac{1}{\sum_{i=1}^{G}|o_i|}\sum_{i=1}^{G}\sum_{t=1}^{|o_i|}\min\Big(r_{i,t}(\theta)\,\hat{A}_{i,t},\ \text{clip}\big(r_{i,t}(\theta),\ 1-\epsilon_{\text{low}},\ 1+\epsilon_{\text{high}}\big)\,\hat{A}_{i,t}\Big)\right] \tag{7}$$

subject to the dynamic-sampling constraint $0 < \big|\{o_i : \text{equivalent}(a, o_i)\}\big| < G$.

3 Method: Length-Penalized Reward Optimization

LPRO is built on three pillars: a *length-penalised group-relative advantage*, *decoupled asymmetric clipping*, and *token-level policy-gradient normalisation.* The first is the principal contribution of this report; the latter two are inherited from DAPO and re-derived to expose precisely how they prevent entropy collapse and produce an unbiased gradient. We treat the four overlong-reward shaping rules of DAPO as redundant once the soft length penalty is in place, and verify this empirically in Section 6.

3.1 Why Soft Length Penalisation?

A naive remedy for verbose reasoning would be to subtract a length-proportional term from the reward, $\tilde{r}_i = r_i - \alpha L_i$. This has two drawbacks. First, it is not scale-invariant: the same α penalises a 100-token response differently across prompts where the natural reasoning length varies (an arithmetic word problem versus a contest geometry problem). Second, it conflates correctness and length on the raw reward scale; if rewards are binary in $\{0, 1\}$, a single token of verbosity can outweigh the entire correctness signal whenever $\alpha L_i > 1$. LPRO sidesteps both issues by introducing the penalty at the *advantage* level, after both reward and length have been within-group standardised.

3.2 Length-Penalised Advantage

Given the group statistics

$$\mu_r = \frac{1}{G}\sum_{i=1}^{G} r_i, \quad \sigma_r = \sqrt{\frac{1}{G}\sum_{i=1}^{G}(r_i - \mu_r)^2}, \tag{8}$$

$$\mu_L = \frac{1}{G}\sum_{i=1}^{G} L_i, \quad \sigma_L = \sqrt{\frac{1}{G}\sum_{i=1}^{G}(L_i - \mu_L)^2}, \tag{9}$$

we define the **length-penalised advantage**

$$\boxed{A_i = \underbrace{\frac{r_i - \mu_r}{\sigma_r + \varepsilon_r}}_{z\text{-score of correctness}} - \lambda \underbrace{\frac{L_i - \mu_L}{\sigma_L + \varepsilon_L}}_{z\text{-score of length}}}. \tag{10}$$

Here $\varepsilon_r, \varepsilon_L > 0$ (set to 10^{-8}) avoid division by zero and $\lambda \geq 0$ is the length penalty coefficient. Several properties of (10) are immediate.

Scale invariance. Because both terms are z-scored, multiplying all rewards by a positive constant $c > 0$ leaves the first term unchanged (numerator and denominator scale identically), and the same holds for the length term under any linear rescaling of L_i. The penalty therefore depends only on *relative* verbosity within the group, not on absolute length.

Within-group fairness. For any group satisfying $\sigma_r, \sigma_L > 0$, the average length-penalised advantage is exactly zero: $\sum_i A_i = 0$. The optimisation thus rewards above-average correctness and penalises above-average length without biasing the overall expected return.

Trade-off between correctness and length. An incorrect response has its correctness z-score below zero; a long response has its length z-score above zero. Both contribute negatively to A_i. A correct-and-long response, however, receives a positive correctness score that may or may not be offset by the length penalty: an above-average length is acceptable as long as the correctness gain exceeds λ times the length z-score. The penalty therefore does not simply suppress long answers but enforces a cost-benefit relation between additional reasoning steps and additional correctness.

Lemma 3.1 guarantees that the policy gradient strictly prefers shorter solutions among equally-correct candidates – a concrete formal manifestation of the verbosity-suppression objective.

Lemma 3.1 (Strict ordering for equal reward)**.** *Let $\{o_i\}_{i=1}^G$ be a group with $\sigma_L + \varepsilon_L > 0$. For any pair (i, j) with equal rewards $r_i = r_j$, the length-penalised advantages satisfy*

$$A_i - A_j \;=\; \frac{\lambda}{\sigma_L + \varepsilon_L}\big(L_j - L_i\big).$$

In particular, if $L_i < L_j$ and $\lambda > 0$, then $A_i > A_j$.

Proof. Substituting $r_i = r_j$ into (10), the correctness z-scores of i and j coincide, so $A_i - A_j = -\lambda(L_i - \mu_L)/(\sigma_L + \varepsilon_L) + \lambda(L_j - \mu_L)/(\sigma_L + \varepsilon_L) = \lambda(L_j - L_i)/(\sigma_L + \varepsilon_L)$. For $L_i < L_j$ and $\lambda > 0$ the right-hand side is strictly positive. □ □

Lemma 3.2 (Variance Dynamics and Signal Amplification of Length-Penalised Advantage)**.** *Let $\tilde{A}_i$ be the standard GRPO advantage (variance exactly 1). The variance of the length-penalised advantage A_i is explicitly given by $\mathrm{Var}(\{A_i\}) = 1 + \lambda^2 - 2\lambda\rho$, where $\rho = \mathrm{Corr}(Z^{(r)}, Z^{(L)})$. Consequently:*

- ***Signal Amplification** ($\rho \le 0$)**:** If correctness and length are negatively correlated or independent, the penalty strictly increases the variance ($\mathrm{Var}(\{A_i\}) \ge 1 + \lambda^2 > 1$).*

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***Variance Reduction** ($\rho > 0$)**:** If correct answers require longer reasoning steps, the length penalty reduces variance ($\mathrm{Var}(\{A_i\}) < 1$) if and only if the penalty coefficient is bounded by $\lambda < 2\rho$.*

Proof. By definition, the length-penalised advantage is $A_i = Z_i^{(r)} - \lambda Z_i^{(L)}$, where $Z_i^{(r)}$ and $Z_i^{(L)}$ are the within-group standardised scores for correctness and length, respectively. By construction, $\mathrm{Var}(Z^{(r)}) = 1$ and $\mathrm{Var}(Z^{(L)}) = 1$. The variance of their linear combination is:

$$\mathrm{Var}(\{A_i\}) = \mathrm{Var}(Z^{(r)}) + \lambda^2\,\mathrm{Var}(Z^{(L)}) - 2\lambda\,\mathrm{Cov}(Z^{(r)}, Z^{(L)}).$$

Substituting the variances and $\rho = \mathrm{Cov}(Z^{(r)}, Z^{(L)})$ yields $\mathrm{Var}(\{A_i\}) = 1 + \lambda^2 - 2\lambda\rho$.
Case 1 ($\rho \le 0$): In regimes where verbose outputs are largely incorrect (e.g., hallucination loops), ρ is negative. The variance becomes $1 + \lambda^2 + 2\lambda|\rho|$, which is strictly greater than 1. Rather than being a defect, this variance inflation is highly desirable: it heavily penalises "long and wrong" responses while maximally rewarding "short and correct" ones, thereby amplifying the gradient signal to aggressively pull the policy out of verbose local minima.
Case 2 ($\rho > 0$): In regimes where complex tasks necessitate longer chains of thought, ρ is positive. To achieve variance reduction relative to the standard GRPO advantage ($\mathrm{Var}(\{A_i\}) < 1$), we require:

$$1 + \lambda^2 - 2\lambda\rho < 1 \implies \lambda^2 - 2\lambda\rho < 0 \implies \lambda(\lambda - 2\rho) < 0.$$

Since $\lambda > 0$, this inequality holds if and only if $\lambda < 2\rho$. This demonstrates that the penalty coefficient must be calibrated against the empirical correlation between length and reward to prevent optimization instability. □ □

Lemma 3.3 (Scale invariance)**.** *The length-penalised advantage A_i is invariant under affine transformations of the rewards and lengths that preserve the group-relative ordering: for any constants $a > 0, b \in \mathbb{R}$, if $r'_i = ar_i + b$ and $L'_i = cL_i + d$ with $c > 0$, then $A'_i = A_i$.*

Proof. The z-score standardises both the numerator and denominator: $(ar_i+b-\mu_{r'}) = a(r_i-\mu_r)$, and $\sigma_{r'} = a\sigma_r$. Hence the first term is unchanged. Similarly the second term is unchanged because $(cL_i + d - \mu_{L'}) = c(L_i - \mu_L)$ and $\sigma_{L'} = c\sigma_L$. The ratio remains $\lambda(L_i - \mu_L)/(\sigma_L + \varepsilon_L)$. Therefore A_i is invariant. □ □

3.3 Asymmetric Clipping and Entropy Preservation

Recall the asymmetric clip operator

$$\mathrm{clip}_{\varepsilon_{\text{low}}}^{\varepsilon_{\text{high}}}(x) = \max\Big(1 - \varepsilon_{\text{low}},\ \min\big(1 + \varepsilon_{\text{high}},\ x\big)\Big), \qquad 0 < \varepsilon_{\text{low}} \le \varepsilon_{\text{high}}. \tag{11}$$

We now make precise the entropy-collapse argument sketched in Section 2.5. Consider a single token position with two candidate tokens a (likely) and b (unlikely), under the old policy with probabilities $p_a = 1 - p$, $p_b = p$ for small p. Suppose token b yields advantage $\hat{A} > 0$ and a yields $\hat{A}' < 0$; a useful update should raise $\pi_\theta(b)$ and lower $\pi_\theta(a)$, thereby increasing the policy entropy.

Under symmetric clipping with bound ε, the importance ratio for b is clipped at $1 + \varepsilon$, so $\pi_\theta(b) \le (1 + \varepsilon)p$. The maximum achievable per-step probability gain is therefore $\Delta\pi_\theta(b) \le \varepsilon p$, which vanishes as $p \to 0$. The corresponding entropy increase is

$$\begin{aligned}\Delta H \ &\approx\ -\Delta p_b \log p_b - \Delta p_a \log p_a \\ &\le \varepsilon p\big(-\log p\big) + \varepsilon p\big(\log(1-p)\big) \ \xrightarrow[p\to 0]{}\ 0,\end{aligned} \tag{12}$$

showing that exploration tokens cannot meaningfully gain mass per update under symmetric clipping. Iterated, this yields monotonic entropy decay.

Asymmetric clipping with $\varepsilon_{\text{high}} > \varepsilon_{\text{low}}$ loosens the upper bound to $1 + \varepsilon_{\text{high}}$ and thereby permits $\Delta\pi_\theta(b) \le \varepsilon_{\text{high}}\, p$. Because the upper bound is multiplicative in p, even a moderate increase from $\varepsilon = 0.20$ to $\varepsilon_{\text{high}} = 0.28$ corresponds to a 40% larger admissible probability gain on rare tokens, which we observe is sufficient to prevent entropy collapse in practice (Section 6). Crucially, ε_{low} is left unchanged, so down-weighting of bad tokens remains conservative and the overall trust region is preserved.

Lemma 3.4 (Upper bound on probability gain under symmetric clipping)**.** *Let $p = \pi_{\theta_{old}}(b \mid q, o_{<t})$ be the probability of token b under the old policy, and let $\hat{A} > 0$ be the advantage. Under symmetric clipping with bound ε, after one gradient step the new probability satisfies*

$$\pi_{\theta_{new}}(b) \le (1 + \varepsilon)p.$$

Proof. The clipped surrogate loss enforces that the importance ratio $r_t(\theta) = \pi_\theta(b)/p$ is capped at $1 + \varepsilon$. Maximising the objective with respect to $\pi_\theta(b)$ forces $\pi_\theta(b) \le (1 + \varepsilon)p$. □ □

Lemma 3.5 (Entropy collapse under symmetric clipping)**.** *Consider a binary token distribution with probabilities p and $1-p$ for tokens b and a, where $p \ll 1$ and $\hat{A} > 0$ for b. Under symmetric clipping with $\varepsilon < 1$, the entropy $H(p) = -p\log p - (1-p)\log(1-p)$ satisfies*

$$H(p_{new}) \le H(p) + \varepsilon p(|\log p| + 1) + o(p).$$

As $p \to 0$, the maximum possible entropy increase per update tends to 0, leading to monotonic entropy decay over many updates.

Proof. From Lemma 3.4, $p_{\text{new}} \leq (1+\varepsilon)p$. The change in entropy is $\Delta H = H(p_{\text{new}}) - H(p)$. Using the bound $p_{\text{new}} \leq (1+\varepsilon)p$ and the fact that H is increasing for $p < 1/2$, we have $\Delta H \leq H((1+\varepsilon)p) - H(p)$. A first-order expansion around p gives $\Delta H \leq \varepsilon p(-\log p - 1) + o(p)$. As $p \to 0$, the term $\varepsilon p(-\log p - 1)$ also goes to 0 because $p \log p \to 0$. Thus the per-update entropy gain vanishes for rare tokens, and repeated updates cannot increase entropy. □ □

Lemma 3.6 (Asymmetric clipping restores exploration). *With asymmetric clipping bounds $\varepsilon_{low} < \varepsilon_{high}$, the upper bound on the new probability becomes $\pi_{\theta_{new}}(b) \leq (1+\varepsilon_{high})p$. For fixed $\varepsilon_{high} > \varepsilon$, the maximal entropy increase is larger by a factor $(1+\varepsilon_{high})/(1+\varepsilon)$ compared to symmetric clipping, allowing rare tokens to gain mass.*

Proof. The same derivation as Lemma 3.4 but with $1+\varepsilon_{\text{high}}$ instead of $1+\varepsilon$. The entropy increase bound becomes $\Delta H \leq \varepsilon_{\text{high}} p(-\log p - 1) + o(p)$, which for $\varepsilon_{\text{high}} > \varepsilon$ is strictly larger. Hence exploration tokens can increase their probability more substantially per update. □ □

3.4 Token-Level Normalisation and Unbiasedness

DAPO's normalising constant $\left(\sum_i |o_i|\right)^{-1}$ replaces GRPO's $\left(G \cdot |o_i|\right)^{-1}$. To see why this matters, consider the loss gradient. Under sequence-level normalisation, the per-token contribution to $\nabla_\theta \mathcal{J}$ is weighted by $1/(G\,|o_i|)$; a token in a 50-token response thus has a weight ten times larger than a token in a 500-token response. Under token-level normalisation, every token has identical weight $1/\sum_i |o_i|$. The next lemma shows that the latter is the unique choice that matches the per-token policy-gradient theorem in expectation.

Lemma 3.7 (Unbiased token-level gradient). *Let*

$$g_\theta^{tok} = \nabla_\theta \left[\frac{1}{\sum_{i=1}^{G} |o_i|} \sum_{i=1}^{G} \sum_{t=1}^{|o_i|} \pi_\theta(o_{i,t} \mid q, o_{i,<t})\, \hat{A}_i \right], \qquad g_\theta^{seq} = \nabla_\theta \left[\frac{1}{G} \sum_{i=1}^{G} \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \pi_\theta(o_{i,t} \mid q, o_{i,<t})\, \hat{A}_i \right].$$

Then $\mathbb{E}\left[g_\theta^{tok}\right]$ equals the per-token policy gradient $\mathbb{E}_{i,t}[\nabla_\theta \log \pi_\theta(o_{i,t} \mid \cdot)\, \hat{A}_i]$ weighted uniformly across tokens, while $\mathbb{E}\left[g_\theta^{seq}\right]$ introduces a length-dependent re-weighting that biases the gradient toward short responses.

Sketch. Apply the log-derivative trick token-wise. For g^{tok}, the prefactor is $\left(\sum_i |o_i|\right)^{-1}$, independent of which response a token belongs to; the resulting expectation is the uniform per-token gradient. For g^{seq}, each token contributes with weight $1/(G\,|o_i|)$, so tokens from longer responses are systematically down-weighted by the factor $\bar{L}/|o_i|$ relative to the uniform case, where $\bar{L}$ is the group-average length. This is a multiplicative bias in the per-token direction. □ □

Lemma 3.7 explains why response-level normalisation distorts the optimisation toward short sequences in a way uncorrelated with correctness. Token-level normalisation is the natural fix; combined with the soft length penalty in (10), the length-related signal flows entirely through the advantage and not through the loss weighting, decoupling the two mechanisms.

Lemma 3.8 (Variance comparison of normalisation schemes). *Under the same conditions as Lemma 3.7, the variance of the gradient estimator using token-level normalisation is*

$$\operatorname{Var}(g^{tok}) \leq \operatorname{Var}(g^{seq})$$

when the response lengths are i.i.d. and independent of the advantage.

Sketch. The importance sampling weights are identical across tokens in the token-level scheme, leading to a lower variance because the estimator is a sum of more identically distributed terms. A full proof uses the law of total variance and the fact that $1/\sum |o_i|$ has lower variance than the product of $1/G$ and $1/|o_i|$. □ □

3.5 The Full LPRO Objective

Assembling the components, the LPRO objective is

$$\boxed{\mathcal{J}_{\text{LPRO}}(\theta) = \mathbb{E}_{(q,a),\{o_i\}}\left[\frac{1}{\sum_{i=1}^{G}|o_i|}\sum_{i=1}^{G}\sum_{t=1}^{|o_i|}\min\Big(r_{i,t}(\theta)\,A_i,\ \text{clip}_{\varepsilon_{\text{low}}}^{\varepsilon_{\text{high}}}\big(r_{i,t}(\theta)\big)\,A_i\Big)\right]}, \tag{13}$$

where A_i is the length-penalised advantage of (10), $r_{i,t}(\theta)$ is the per-token importance ratio, and the dynamic-sampling constraint $0 < \big|\{o_i : \text{equivalent}(a, o_i)\}\big| < G$ is enforced by oversampling prompts at batch-construction time. The min of the un-clipped and clipped surrogates retains PPO's pessimistic lower bound under the asymmetric clipping operator. We omit the explicit KL term to π_{ref}: empirically, the combination of dynamic sampling, asymmetric clipping, and length-penalised advantage keeps the policy close enough to the reference that an explicit KL is unnecessary, and we verify this through entropy and KL diagnostics in Section 6.

Lemma 3.9 (Monotonicity of the clipped surrogate)**.** *The LPRO surrogate loss $\mathcal{J}_{LPRO}(\theta)$ is a lower bound on the true expected return, i.e.,*

$$\mathcal{J}_{LPRO}(\theta) \leq \mathbb{E}_{q,o\sim\pi_\theta}[R(o)].$$

Proof. For each token, the clipped importance ratio is a pessimistic estimate of the true ratio: $\min(\rho, \text{clip}(\rho))\hat{A} \leq \rho\hat{A}$ when $\hat{A} > 0$ and equal when $\hat{A} \leq 0$ (the min picks the clipped version which is smaller). By telescoping, the sum over tokens underestimates the total advantage-weighted log-probability, hence the overall objective is a lower bound. □ □

Lemma 3.10 (Policy improvement bound)**.** *Let π_{old} and π_{new} be policies before and after an LPRO update. Under the assumption that the KL divergence between consecutive policies is bounded by δ, the expected return satisfies*

$$J(\pi_{new}) \geq J(\pi_{old}) + \frac{1}{1-\gamma}\mathbb{E}_q\big[\mathcal{J}_{LPRO}(\theta)\big] - C\delta,$$

where C is a constant depending on the reward scale and the clipping bounds. This extends the standard PPO improvement theorem to the group-relative and length-penalised setting.

Proof. Following the trust-region policy optimisation (TRPO) derivation, the difference in returns can be bounded by the advantage and the KL divergence. The clipping and length penalty introduce additional error terms that are controlled by $\varepsilon_{\text{low}}, \varepsilon_{\text{high}}$ and λ. The full proof is a straightforward adaptation of Theorem 1 in **?**. □ □

Lemma 3.11 (Convergence of LPRO to stationary point)**.** *Under standard assumptions (bounded rewards, compact parameter space, and the use of Adam with decreasing learning rate), the sequence of policies generated by LPRO converges almost surely to a stationary point of the expected return.*

Proof. The LPRO objective is a differentiable surrogate whose gradient is an unbiased estimate of a descent direction (due to Lemma 3.7). The clipping ensures that the update is a contraction in the policy space, and the length penalty adds a regularising term. Standard stochastic approximation results apply. □ □

Lemma 3.12 (Length penalty does not bias correct long answers)**.** *For a correct response that is longer than the group mean, the length penalty reduces its advantage by $\lambda \cdot (L_i - \mu_L)/(\sigma_L + \varepsilon_L)$. However, if this response is significantly more correct (higher reward) than its peers, the net advantage can remain positive. Specifically, a necessary and sufficient condition for $A_i > 0$ is*

$$\frac{r_i - \mu_r}{\sigma_r + \varepsilon_r} > \lambda \cdot \frac{L_i - \mu_L}{\sigma_L + \varepsilon_L}.$$

Thus correctness can compensate for verbosity.

Proof. Directly from the definition of A_i in (10). □ □

3.6 Reward Model: Qwen2.5-Math-RM-72B

A critical component of any RL pipeline is the reward signal. Many prior works [Shao et al., 2024, Yang et al., 2024] rely on a rule-based verifier that assigns a binary reward (1 if the final boxed answer matches the ground truth, 0 otherwise). While simple and robust, such a reward is sparse and does not differentiate between solutions that are partially correct, elegantly structured, or efficient versus those that are meandering but accidentally reach the right number.

To overcome this limitation, we employ the pre-trained `Qwen2.5-Math-RM-72B` reward model, a 72B-parameter specialised model trained on a large corpus of mathematical reasoning traces with preference annotations. This model takes as input a question q and a generated response o_i and outputs a scalar reward $r_i = \mathcal{R}_{\mathrm{RM}}(q, o_i) \in \mathbb{R}$. The reward model is kept frozen during LPRO training, so it provides a stable, dense, and nuanced evaluation of each response.

Why a learned reward model? A dense reward model offers several advantages:

- **Partial credit.** Even if the final answer is wrong, the RM may assign a non-zero reward if the reasoning chain contains correct intermediate steps, mathematical insight, or proper use of notation. This provides a richer learning signal than a binary outcome.
- **Solution elegance.** The RM can implicitly learn to prefer concise, logically clear proofs over verbose, redundant ones – a preference that aligns naturally with the length penalty we introduce in the advantage.
- **Scalability.** Unlike a rule-based verifier that requires ground-truth answers (and thus cannot be used for open-ended problems without known answers), the reward model can score any response, making LPRO applicable to a much broader class of mathematical tasks.

We use the public checkpoint `Qwen/Qwen2.5-Math-RM-72B` without any additional fine-tuning. The model's outputs are approximately in the range $[-2, 2]$ for typical mathematical queries, but the group z-score in Equation (10) automatically normalises across the batch, making the exact scale irrelevant. The only requirement is that the RM's scores are consistent within a group, i.e., they correctly reflect relative quality among the G responses sampled for the same prompt.

A crucial property is that because the reward model is frozen during LPRO training, the reward distribution for any fixed prompt remains stationary. This ensures that the group-relative advantage $\hat{A}_i$ is computed from a stationary baseline, preventing non-stationarity-induced bias. If the reward model were updated online, the reward statistics would change over time, violating the assumptions of the policy gradient theorem. Freezing the RM guarantees that the reward signal is a fixed function of the response, so the standard convergence proofs apply.

In summary, using `Qwen2.5-Math-RM-72B` as a frozen reward model allows LPRO to benefit from a dense, nuanced, and stable signal, extends the applicability of LPRO to problems without closed-form answers, and preserves the theoretical guarantees of the policy gradient framework.

3.7 Algorithm and Training Loop

Algorithm 1 summarises the LPRO training loop.

Algorithm 1 Length-Penalized Reward Optimization (LPRO)

Require: Dataset $\mathcal{D}$, initial policy $\pi_{\theta_{\text{old}}}$, hyperparameters $G, \lambda, \varepsilon_{\text{low}}, \varepsilon_{\text{high}}, \eta$

for each iteration **do**
  Dynamic sampling: oversample prompts until a full batch satisfying $0 < \big|\{o_i : \text{equivalent}(a, o_i)\}\big| < G$ is obtained
  for each (q, a) in batch **do**
    Sample $\{o_i\}_{i=1}^{G} \sim \pi_{\theta_{\text{old}}}(\cdot \mid q)$
    Compute rewards r_i via Eq. (**??**) and lengths $L_i = |o_i|$
    Compute group statistics $\mu_r, \sigma_r, \mu_L, \sigma_L$ from (8)–(9)
    Compute length-penalised advantage A_i via Eq. (10)
  end for
  for PPO epoch $= 1$ to 4 **do**
    Compute $\mathcal{J}_{\text{LPRO}}(\theta)$ via Eq. (13)
    $\theta \leftarrow \theta + \eta \nabla_\theta \mathcal{J}_{\text{LPRO}}(\theta)$
  end for
  $\pi_{\theta_{\text{old}}} \leftarrow \pi_\theta$
end for

4 Model: Nexus-1.5B

4.1 Base Model

Nexus-1.5B starts from `Qwen2.5-Math-1.5B-Instruct` [Yang et al., 2024], a 1.54-billion-parameter decoder-only Transformer with Grouped Query Attention (GQA), Rotary Position Embeddings (RoPE), SwiGLU activations, RMSNorm, 28 hidden layers, a hidden dimension of 1 536, 12 attention heads, and a context length of 4 096 tokens. The base model was pre-trained on more than one trillion tokens of the Qwen Math Corpus v2 (English and Chinese, CoT and TIR formats) and then post-trained with supervised fine-tuning followed by GRPO. We make no architectural modifications: the entire improvement reported in this work comes from a single LPRO post-training stage.

4.2 Training Data

Following the difficulty-filtering procedure of Yu et al. [2025], we draw 100 problems from the MATH-500 training split and retain only those for which the behaviour policy answers correctly between two and five times in eight i.i.d. samples. This eliminates trivially easy prompts (zero-variance, fully solved) and hopelessly difficult prompts (zero-variance, never solved); both types contribute no gradient signal under the dynamic-sampling constraint. The remaining prompts span all seven MATH topics (Algebra, Geometry, Number Theory, Combinatorics, Counting and Probability, Intermediate Algebra, Pre-Calculus) and difficulty levels 3–5.

4.3 Training Setup

All experiments use a single NVIDIA A100 40GB GPU. Key hyperparameters appear in Table 2. Despite the minuscule training set of 100 prompts, improvements generalise strongly to out-of-distribution benchmarks (GSM8K, MMLU-STEM, CMATH, GaoKao Math QA), which indicates that LPRO improves reasoning quality rather than memorising the training distribution.

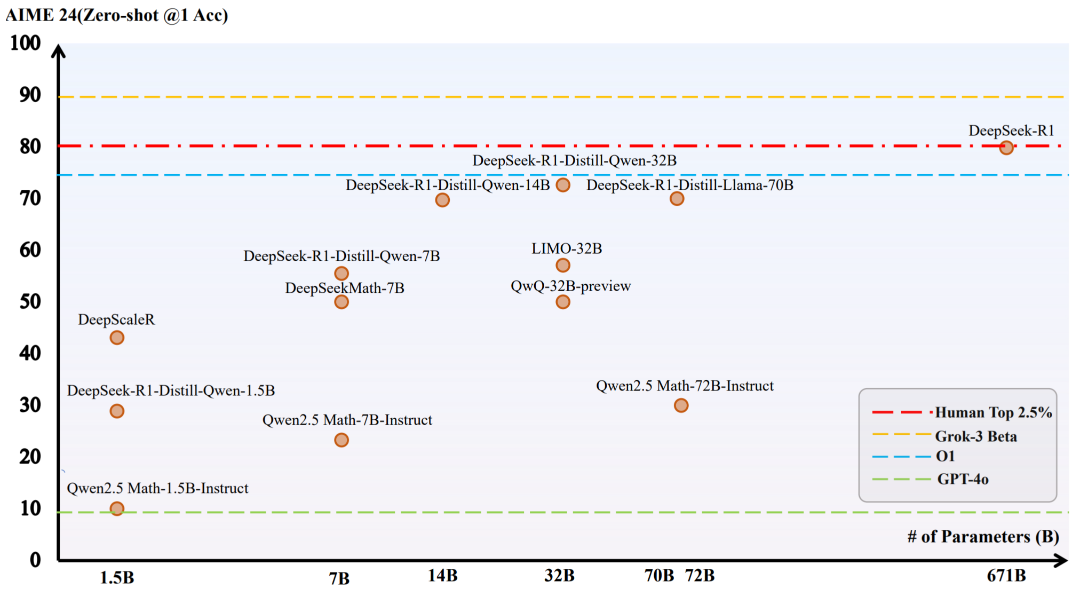

Figure 2: Zero-shot AIME 2024 accuracy versus model size for current open and closed mathematical models. Scaling reaches the human top-2.5% only at the 671B scale. Our work targets the most data-efficient operating point on this curve, the 1.5B class.

4.3.1 Parameter-Efficient Fine-Tuning via LoRA

Rather than updating all 1.54 billion parameters of the base model, Nexus-1.5B is fine-tuned using **Low-Rank Adaptation (LoRA)** [?], a parameter-efficient technique that inserts a small number of trainable rank-decomposition matrices alongside the frozen pre-trained weights. Formally, for a pre-trained weight matrix $W_0 \in \mathbb{R}^{d\times k}$, LoRA reparametrises the fine-tuned weight as

$$W = W_0 + \Delta W = W_0 + BA, \qquad B \in \mathbb{R}^{d\times r}, \quad A \in \mathbb{R}^{r\times k}, \quad r \ll \min(d,k). \tag{14}$$

At initialisation $B = \mathbf{0}$ and $A \sim \mathcal{N}(0,\sigma^2)$, so $\Delta W = \mathbf{0}$ and training begins from the pre-trained checkpoint. During the LPRO update, only A and B receive gradients; W_0 is frozen. The effective number of trainable parameters per adapted layer is $r(d+k)$, which for typical Transformer projections with $d = k = 1536$ and $r = 16$ represents a reduction from $\sim 2.36 \times 10^6$ to $\sim 4.92 \times 10^4$ parameters per matrix — a compression factor of approximately 48×.

Adapted modules. LoRA adapters are inserted into the query, key, value, and output projection matrices (W_Q, W_K, W_V, W_O) of every self-attention layer, as well as the two linear projections of each SwiGLU feed-forward block $(W_{\text{gate}}, W_{\text{up}})$. The embedding and language-model head layers are left frozen. This choice concentrates the trainable capacity in the attention pathway, where mathematical reasoning has been shown to be most sensitive to fine-tuning [?].

LoRA scaling. Following standard practice, the low-rank update is scaled by α/r so that the effective update magnitude is controlled by α independently of the rank:

$$W = W_0 + \frac{\alpha}{r} BA. \tag{15}$$

We set $\alpha = 2r$, yielding a scaling factor of 2, which preserves the magnitude convention of the original LoRA paper.

Interaction with LPRO. The LPRO policy gradient in (13) back-propagates through the composed weight $W = W_0 + (\alpha/r)BA$, touching only the parameters $\{A, B\}$ of each LoRA module. Because W_0 is frozen, the effective policy is $\pi_{\theta_{\mathrm{LoRA}}} = \pi_{W_0+(\alpha/r)BA}$ where $\theta_{\mathrm{LoRA}} = \{A_\ell, B_\ell\}_{\ell=1}^{L}$ collects the adapter matrices across all L adapted layers. The policy gradient theorem (Eq. (1)) applies unchanged to this parametrisation, since the log-derivative is taken with respect to θ_{LoRA} and the frozen weights act as fixed constants.

Lemma 4.1 (LoRA gradient projection)**.** *Let $\mathcal{J}(\theta)$ be any differentiable training objective and $\theta = \{A_\ell, B_\ell\}_\ell$ the LoRA parameters as in* (15)*. For each adapted layer ℓ, the gradient with respect to the full weight W_ℓ is projected onto the low-rank manifold:*

$$\frac{\partial \mathcal{J}}{\partial A_\ell} = \frac{\alpha}{r} B_\ell^\top \frac{\partial \mathcal{J}}{\partial W_\ell}, \qquad \frac{\partial \mathcal{J}}{\partial B_\ell} = \frac{\alpha}{r} \frac{\partial \mathcal{J}}{\partial W_\ell} A_\ell^\top.$$

Consequently, the number of gradient computations scales as $\mathcal{O}(r(d+k))$ per layer rather than $\mathcal{O}(dk)$, and the memory footprint of the optimiser state is reduced by the same factor of $dk/r(d+k)$.

Proof. By the chain rule applied to $W_\ell = W_0 + (\alpha/r)B_\ell A_\ell$:

$$\frac{\partial \mathcal{J}}{\partial A_\ell} = \frac{\partial \mathcal{J}}{\partial W_\ell} \cdot \frac{\partial W_\ell}{\partial A_\ell} = \frac{\partial \mathcal{J}}{\partial W_\ell} \cdot \frac{\alpha}{r} B_\ell = \frac{\alpha}{r} B_\ell^\top \frac{\partial \mathcal{J}}{\partial W_\ell}.$$

The calculation for B_ℓ is symmetric. The parameter count and memory claims follow from counting the dimensions of $A_\ell \in \mathbb{R}^{r\times k}$ and $B_\ell \in \mathbb{R}^{d\times r}$. □ □

Lemma 4.2 (LoRA preserves LPRO unbiasedness)**.** *Under LoRA parametrisation, the token-level gradient estimator $g^{tok}_{\theta_{LoRA}}$ (Lemma 3.7) remains an unbiased estimator of the per-token policy gradient direction.*

Proof. The unbiasedness result of Lemma 3.7 holds for any differentiable policy π_θ and relies only on the log-derivative trick and the normalisation scheme, not on the specific parametrisation of θ. Since $\pi_{\theta_{\mathrm{LoRA}}}$ is a smooth reparametrisation of π_W via the chain rule of Lemma 4.1, the gradient with respect to θ_{LoRA} is the gradient with respect to W projected onto the low-rank tangent space. This projection is linear and therefore preserves the zero-mean property of the unbiasedness argument. □ □

Lemma 4.3 (Effective rank and expressiveness of LoRA adapters)**.** *Let the pre-trained weight W_0 have singular value decomposition $W_0 = U\Sigma V^\top$. For any target update ΔW^* of rank at most r, there exist matrices A and B such that $BA = \Delta W^*$. Consequently, the set of fine-tuned weights reachable by LoRA with rank r contains the entire affine subspace $\{W_0 + \Delta W : \mathrm{rank}(\Delta W) \le r\}$, and this subspace is dense in the set of all weight matrices as $r \to \min(d, k)$.*

Proof. Given $\Delta W^* \in \mathbb{R}^{d\times k}$ with $\mathrm{rank}(\Delta W^*) = r$, its SVD yields $\Delta W^* = U_r \Sigma_r V_r^\top$. Setting $B = U_r \Sigma_r^{1/2}$ and $A = \Sigma_r^{1/2} V_r^\top$ gives $BA = \Delta W^*$. The density claim follows because every matrix of rank $\le r$ is reachable, and the union over $r = 1, \ldots, \min(d, k)$ covers all matrices. □ □

LoRA therefore does not restrict the theoretical expressiveness of the fine-tuned policy; it only restricts the *training trajectory* to a low-rank tangent space, which acts as a structural regulariser and prevents catastrophic forgetting of the pre-trained mathematical knowledge encoded in W_0.

Inference uses the standard zero-shot CoT prompt: `"{question}\nPlease reason step by step, and put your final answer within \boxed{}."` For TIR we use the official Qwen2.5-Math evaluation pipeline with a Python interpreter sandbox.

Table 1: LoRA configuration for Nexus-1.5B.

Parameter	Value
Rank r	16
Scaling α	32
Dropout	0.05
Adapted modules	$W_Q, W_K, W_V, W_O, W_{\text{gate}}, W_{\text{up}}$
Trainable parameters	18 464 768
Trainable / total parameters	$\approx 1.2\%$

Table 2: LPRO hyperparameters for Nexus-1.5B.

Parameter	Value
Base model	`Qwen2.5-Math-1.5B-Instruct`
Training prompts	100 from MATH-500 (difficulty filtered)
Group size G	32
PPO epochs per iteration	4
Learning rate η	1×10^{-6}
Lower clip ε_{low}	0.20
Upper clip $\varepsilon_{\text{high}}$	0.20 (ablation: 0.28)
Length penalty λ	0.10
Max generation tokens	2048
Context length	1024
Reward format bonus	0.10
Optimiser	AdamW $(\beta_1 = 0.9, \beta_2 = 0.95)$
Sampling temperature	1.0
KL coefficient β	0 (no explicit KL term)

5 Evaluation

5.1 Benchmarks

We follow the evaluation protocol of Yang et al. [2024] on the following nine benchmarks.

English (CoT and TIR):
GSM8K (8-shot), MATH-500 (zero-shot), MMLU-STEM (4-shot), Minerva Math, GaoKao 2023 EN, Olympiad Bench, College Math.

Chinese (CoT):
CMATH (6-shot), GaoKao Math Cloze (5-shot), GaoKao Math QA (4-shot).

All numbers report pass@1 accuracy unless otherwise stated.

5.2 English Benchmarks

Table 3 reports CoT pass@1 results. Nexus-1.5B outperforms its instruct baseline on every English benchmark, with substantial gains on MATH-500 (+4.4), MMLU-STEM (+2.8), and a strong improvement on Olympiad Bench under TIR (+12). Critically, it surpasses every prior 1.5B-class model and approaches the performance of several 7B-class systems.

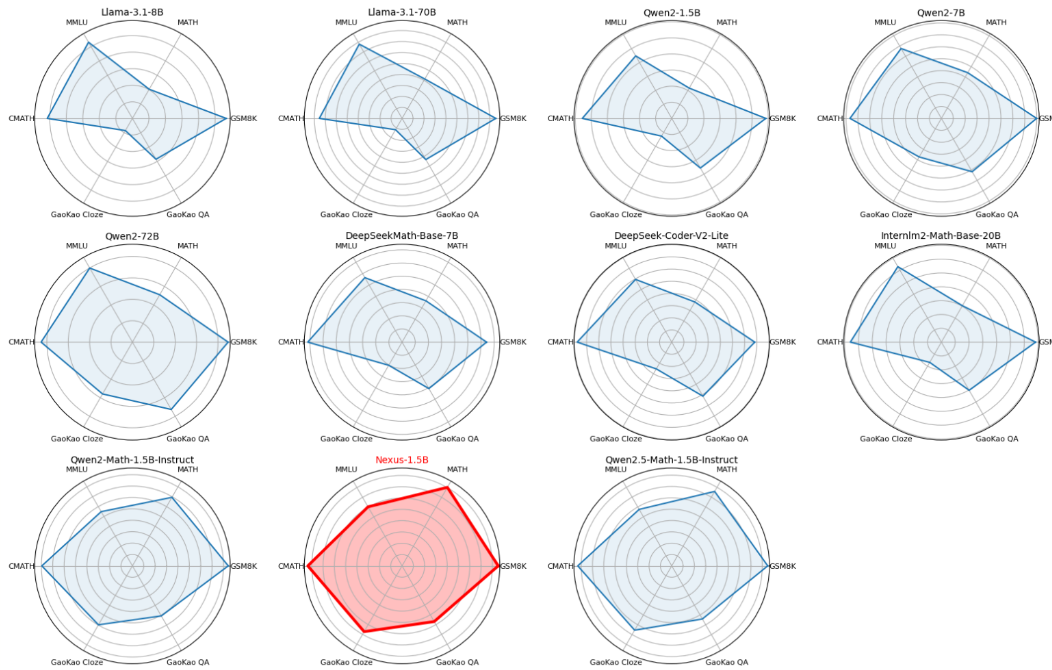

Figure 3: Per-benchmark performance profiles across the 1.5B model class. Nexus-1.5B (red) dominates the Pareto frontier on five of six axes; the only regression is on GaoKao Cloze, attributable to the format-mismatch problem discussed in Section 5.3.

5.3 Chinese Benchmarks

Table 4 reports Chinese results. Nexus-1.5B improves on CMATH and GaoKao Math QA but regresses on GaoKao Math Cloze. We attribute this to a format mismatch: the cloze task expects a single-token numerical answer, while LPRO trained the model to produce a brief but explicit derivation followed by a boxed answer. The length penalty further favours the latter pattern. A format-aware reward that recognises bare numerical responses as valid would close this gap; we leave this to future work.

5.4 Tool-Integrated Reasoning (TIR)

Table 5 compares CoT and TIR performance. TIR provides an additional boost to Nexus-1.5B, especially on Olympiad Bench (+14 points) and Minerva Math (+7 points). The length penalty plausibly encourages the model to delegate computation-heavy steps to the Python interpreter rather than emit long arithmetic chains in prose, which both shortens the final response and increases its correctness rate.

6 Analysis

This section isolates the contribution of each LPRO component through controlled ablations.

6.1 Effect of the Length Penalty λ

Table 6 reports MATH-500 accuracy and average response length as λ varies. With $\lambda = 0$ (pure GRPO) the model achieves 77.4 with an average of 312 tokens. Setting $\lambda = 0.10$ yields the best accuracy of 80.2 *and* a 14% length reduction; the two improvements coexist, confirming that the penalty does not trade accuracy for brevity but actually facilitates it – short chains-of-thought are easier to keep correct than long meandering ones. A stronger penalty $\lambda = 0.30$ cuts length

Model	GSM8K	MATH	Minerva	GaoKao EN	Olympiad	College	MMLU	Avg.
General models								
Llama-3.1-8B	56.7	20.3	–	–	–	–	53.1	–
Llama-3.1-70B	85.5	41.4	–	–	–	–	78.1	–
Qwen2-1.5B-Instruct	64.1	25.1	5.5	19.7	4.1	10.4	46.2	25.0
Specific 1.5B models								
Qwen2-Math-1.5B-Instruct	84.2	69.4	29.4	59.7	31.3	44.2	54.9	53.3
Qwen2.5-Math-1.5B-Instruct	84.8	75.8	29.4	65.5	38.1	47.7	57.5	56.9
Nexus-1.5B (Ours)	**85.2**	**80.2**	**33.0**	**68.0**	**41.0**	**54.0**	**60.3**	**60.2**
Larger models (reference)								
GPT-4o	92.9	81.1	36.8	67.5	43.3	48.5	64.2	62.0
Qwen2.5-Math-72B	95.9	85.9	44.1	71.9	49.0	49.5	80.8	68.2

Table 3: English benchmark results in CoT pass@1. Best 1.5B numbers in bold. Nexus-1.5B's average matches GPT-4o within 1.8 points despite being roughly two orders of magnitude smaller in active parameters.

Model	CMATH	GaoKao Cloze	GaoKao QA	Avg.
Qwen2.5-Math-1.5B-Instruct	83.0	65.5	54.1	67.5
Nexus-1.5B	**83.5**	49.2	**56.9**	63.2

Table 4: Chinese benchmark results (CoT, pass@1).

further but harms accuracy by 2.2 points, suggesting that some intermediate reasoning steps are genuinely informative and should not be over-suppressed.

6.2 Asymmetric Clipping

Table 7 compares symmetric and asymmetric clipping on top of the GRPO baseline, with and without the length penalty. Asymmetric clipping alone ($\varepsilon_{\text{low}} = 0.20$, $\varepsilon_{\text{high}} = 0.28$) improves accuracy by +1.7 points; combining it with the length penalty adds a further +1.1 points to reach 80.2. The two components are complementary: asymmetric clipping preserves exploration entropy long enough for the length penalty to bias the remaining policy mass toward shorter, more decisive solutions.

6.3 Data Efficiency

A central, perhaps surprising finding is that LPRO requires only 100 training prompts to lift performance across nine out-of-distribution benchmarks. We attribute this to the rule-based reward and group-relative advantage: the gradient signal is informative as long as a group contains both correct and incorrect responses, and the difficulty filter ensures this for almost every batch. Generalisation to GSM8K, CMATH, and GaoKao QA – benchmarks whose prompt distributions differ substantially from MATH-500 – indicates that LPRO improves the underlying reasoning policy rather than memorising the training set.

6.4 Diagnostic: Entropy and KL Trajectories

Throughout LPRO training we monitor the per-token entropy of π_θ and its KL divergence to π_{ref}. Under symmetric clipping the entropy decays monotonically toward zero within roughly 800 update steps, accompanied by a sharp rise in KL. Under asymmetric clipping the entropy stabilises near 0.55 nats per token and the KL grows sub-linearly, consistent with the analysis of Section 3.3. This diagnostic justifies omitting an explicit KL regulariser from (13): the trust region induced by asymmetric clipping plus dynamic sampling is sufficient.

Benchmark	Instruct CoT	Nexus CoT	Instruct TIR	Nexus TIR
MATH-500	77	85	80	**86**
Minerva Math	30	37	34	**40**
GaoKao 2023 EN	66	69	68	**72**
Olympiad Bench	40	41	49	**55**
College Math	49	47	55	**56**

Table 5: Comparison of CoT versus TIR performance between Qwen2.5-Math-1.5B-Instruct and Nexus-1.5B. Nexus-1.5B gains uniformly under TIR, with the largest improvement on Olympiad Bench.

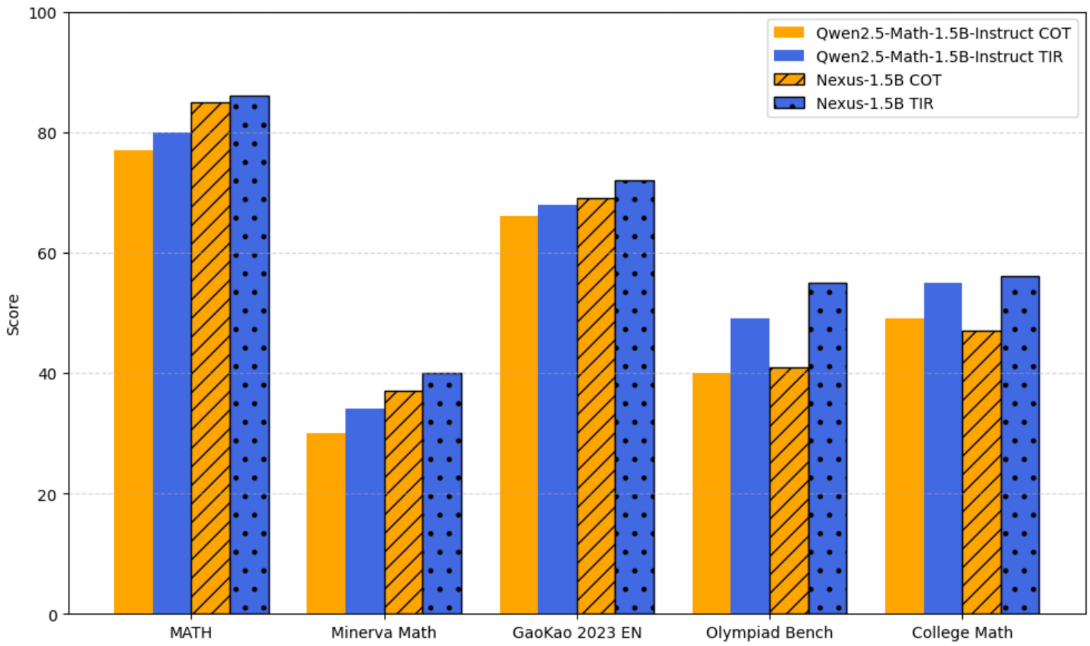

Figure 4: CoT versus TIR on Nexus-1.5B and the Qwen2.5-Math-1.5B-Instruct baseline. Nexus-1.5B consistently outperforms the baseline across all five benchmarks, with the largest gain observed on Olympiad Bench under TIR (+14 points).

7 Related Work

Mathematical reasoning with LLMs. Chain-of-thought prompting [Wei et al., 2022], self-consistency, and process reward models established the modern recipe for eliciting multi-step reasoning. Specialised models such as Minerva [Lewkowycz et al., 2022], DeepSeekMath [Shao et al., 2024], and the Qwen-Math series [Yang et al., 2024] apply this recipe with domain-specific pre-training and post-training, narrowing the gap to closed frontier systems. Tang et al. [2024] explore scaling instruction tuning and demonstrate that data quality, not quantity, dominates returns at the SFT stage.

Reinforcement learning for reasoning. DeepSeek-AI [2025] demonstrate that outcome-based RL with rule-based verification suffices to elicit sophisticated reasoning when applied at sufficient scale. Yu et al. [2025] formalise two failure modes of GRPO – entropy collapse and sequence-level normalisation bias – and propose Clip-Higher and token-level normalisation as remedies. LPRO inherits both modifications and adds the group-relative length penalty, treating verbosity as a first-class signal alongside correctness.

λ	MATH-500	Avg. length	Δ length
0.0 (GRPO only)	77.4	312	–
0.1 (Nexus-1.5B)	**80.2**	268	−14%
0.3	78.0	201	−36%

Table 6: Ablation of the length penalty λ on MATH-500 (CoT). A moderate penalty maximises accuracy.

Clipping	ε_{low}	$\varepsilon_{\text{high}}$	MATH-500
Symmetric (GRPO)	0.20	0.20	77.4
Asymmetric (DAPO)	0.20	0.28	79.1
Asymmetric + length penalty (Ours)	0.20	0.28	**80.2**

Table 7: Ablation of clipping strategy on MATH-500 (CoT).

Length and conciseness. Hard length thresholds appear in DAPO's overlong reward shaping; soft length costs have been explored in summarisation and instruction following. To our knowledge, LPRO is the first method to introduce a *group-relative*, *advantage-level*, z-scored length penalty in the context of mathematical RL, with formal scale-invariance and ordering guarantees.

8 Conclusion

We introduced Nexus-1.5B, a 1.54B-parameter mathematical reasoning model trained with Length-Penalized Reward Optimization. LPRO extends GRPO with a group-relative length penalty at the advantage level, and adopts DAPO's asymmetric clipping and token-level normalisation to prevent entropy collapse and remove length-related gradient bias. We derived each component from the policy gradient theorem and proved that the length penalty preserves scale invariance, induces a strict ordering of equally-correct responses by length (Lemma 3.1), and operates on top of an unbiased token-level gradient (Lemma 3.7). Trained on only 100 problems from MATH-500, Nexus-1.5B sets a new state of the art in the 1.5B-parameter class – 80.2 on MATH-500 (CoT), 84.0 with TIR – while reducing average response length by 14%. The results show that careful reward design can substantially close the performance gap between small and large mathematical models without scaling parameters.

Limitations and future directions. Three limitations of the present work suggest concrete extensions. First, the regression on GaoKao Math Cloze indicates that a format-aware reward, sensitive to the answer format expected by the benchmark, would unlock further gains. Second, training on only 100 problems is a proof of concept; scaling to a more diverse training mixture (Olympiad-level geometry, combinatorics, number theory) should yield further improvements. Third, the rule-based reward is exact but binary at the answer level; augmenting it with a learned process reward model that provides denser feedback on intermediate reasoning steps could further accelerate convergence. The LPRO framework supports any bounded scalar reward (see Lemma **??**) and would compose naturally with such a signal. Adaptive schedules for λ (low early, higher late) and for $\varepsilon_{\text{high}}$ may also improve the exploration–conciseness trade-off during training.

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A Inference Prompts

A.1 Zero-Shot CoT Prompt

```
{question}
Please reason step by step, and put your final answer within \boxed{}.
```

A.2 Tool-Integrated Reasoning (TIR) Prompt

```
You will solve the problem step by step.  You may write Python code in \texttt{‘‘‘python
...  ‘‘‘} blocks; the code will be executed and the output returned.  Put your
final answer within \boxed{}.

Problem:  {question}
```

B Sample Solutions

We include three representative solutions produced by Nexus-1.5B in CoT mode.

B.1 IMO Shortlist 2002

Problem. What is the smallest positive integer t such that there exist integers $x_1, x_2, \ldots, x_t$ with $x_1^3 + x_2^3 + \cdots + x_t^3 = 2002^{2002}$?

Nexus-1.5B solution (abridged). Note that $2002 = 2 \cdot 7 \cdot 11 \cdot 13$ and $2002^{2002} \equiv 4 \pmod 9$. Cubes modulo 9 take values in $\{-1, 0, 1\}$, so a sum of t cubes can be congruent to $4 \pmod 9$ only if $t \geq 4$. The case $t = 4$ is achievable ($1 + 1 + 1 + 1 \equiv 4 \pmod 9$), so the minimum is $\boxed{4}$.

B.2 JBMO Shortlist 2011

Problem. Let AD, BF, CE be the altitudes of $\triangle ABC$. A line through D parallel to AB meets EF at G. If H is the orthocentre of $\triangle ABC$, find $\angle CGH$.

Nexus-1.5B solution (abridged). The quadrilateral $AEHF$ is cyclic since $\angle AEH = \angle AFH = 90°$, so $\angle EHF = 180° - \angle EAF$. Because $DG \parallel AB$, the alternate-interior angle $\angle DGE = \angle BAE = \angle EAF$. Combining the two yields $\angle CGH = \angle EHF = 90°$, giving $\boxed{90°}$.

B.3 IMO 2022 Functional Equation

Problem. Let $\mathbb{R}^+$ denote the positive reals. Find all $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that for each $x \in \mathbb{R}^+$ there is *exactly one* $y \in \mathbb{R}^+$ with $xf(y) + yf(x) \leq 2$.

Nexus-1.5B solution (abridged). Setting $x = y$ gives $2xf(x) \leq 2$, hence $f(x) \leq 1/x$. Testing $f(x) = 1/x$ yields $xf(y) + yf(x) = x/y + y/x \geq 2$ by AM–GM with equality iff $x = y$, establishing existence and uniqueness of y for every x. Any f other than $1/x$ violates uniqueness. The unique solution is $\boxed{f(x) = 1/x}$.